DAV Assignment 1

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In [1]:
```
import pandas as pd
import xml.etree.ElementTree as ET
import json
```

1. Data Collection

Txt File

In [2]:
```
with open("data/students.txt","w") as f:
    f.write("ID Name Age\n")
    f.write("1 Saptanshu 22\n")
    f.write("2 Hardik 21\n")
    f.write("3 Saumya 22")
    f.write("4 Piyush 22")
```

JSON File

In [3]:
```
students = [
    {"id": 1, "name": "Saptanshu", "age": 22},
    {"id": 2, "name": "Hardik", "age": 21},
    {"id": 3, "name": "Saumya", "age": 22},
    {"id": 4, "name": "Piyush", "age": 22}
]

with open("data/students.json","w") as f:
    json.dump(students, f, indent=4)
```

XML File

In [4]:
```
root = ET.Element("students")

for student in students:
    s = ET.SubElement(root, "student")
    ET.SubElement(s, "id").text = str(student["id"])
    ET.SubElement(s, "name").text = student["name"]
```

```
    ET.SubElement(s, "age").text = str(student["age"])

tree = ET.ElementTree(root)
tree.write("data/students.xml")
```

2. CSV Data collection

In [5]:
```
data={
        "ID" :[1,2,3,4],
        "Name": ["Saptanshu","Hardik","Saumya","Piyush"],
        "Age":[22,21,22,22]
        }
df=pd.DataFrame(data)
```

In [6]:
```
df.to_csv("data/students.csv",index=False)
```

In [7]:
```
df1=pd.read_csv("data/students.csv")
print("\nHead")
print(df1)
print("\nTail")
print(df1)
print("\nInfo")
df1.info()
print("\nDescribe")
df1.describe()
```

```
Head
   ID       Name  Age
0   1  Saptanshu   22
1   2     Hardik   21
2   3     Saumya   22
3   4     Piyush   22

Tail
   ID       Name  Age
0   1  Saptanshu   22
1   2     Hardik   21
2   3     Saumya   22
3   4     Piyush   22

Info
<class 'pandas.DataFrame'>
RangeIndex: 4 entries, 0 to 3
Data columns (total 3 columns):
 #   Column  Non-Null Count  Dtype
---  ------  --------------  -----
 0   ID      4 non-null      int64
 1   Name    4 non-null      str
 2   Age     4 non-null      int64
dtypes: int64(2), str(1)
memory usage: 228.0 bytes

Describe
```

Out[7]:

	ID	Age
count	4.000000	4.00
mean	2.500000	21.75
std	1.290994	0.50
min	1.000000	21.00
25%	1.750000	21.75
50%	2.500000	22.00
75%	3.250000	22.00
max	4.000000	22.00

3. Data Analysis

In [8]:
```
customers = pd.DataFrame({
    "CustomerID": [101, 102, 103, 104],
    "Name": ["Saptanshu", "Hardik", None, "Saumya"],
    "Age": [22, 21, None, 22],
    "City": ["Delhi", "Mumbai", "Delhi", None]
})
```

In [9]:
```
orders = pd.DataFrame({
    "OrderID": [1, 2, 3, 4],
    "CustomerID": [101, 102, 101, 104],
    "Amount": [500, 700, None, 450]
})
```

In [10]:
```
print("\nCustomers Data")
print(customers)

print("\nOrders Data")
print(orders)

print("\nCustomers Info")
customers.info()

print("\nOrders Info")
orders.info()
```

```
Customers Data
   CustomerID       Name   Age    City
0         101  Saptanshu  22.0   Delhi
1         102     Hardik  21.0  Mumbai
2         103        NaN   NaN   Delhi
3         104     Saumya  22.0     NaN

Orders Data
   OrderID  CustomerID  Amount
0        1         101   500.0
1        2         102   700.0
2        3         101     NaN
3        4         104   450.0

Customers Info
<class 'pandas.DataFrame'>
RangeIndex: 4 entries, 0 to 3
Data columns (total 4 columns):
 #   Column      Non-Null Count  Dtype
---  ------      --------------  -----
 0   CustomerID  4 non-null      int64
 1   Name        3 non-null      str
 2   Age         3 non-null      float64
 3   City        3 non-null      str
dtypes: float64(1), int64(1), str(2)
memory usage: 260.0 bytes

Orders Info
<class 'pandas.DataFrame'>
RangeIndex: 4 entries, 0 to 3
Data columns (total 3 columns):
 #   Column      Non-Null Count  Dtype
---  ------      --------------  -----
 0   OrderID     4 non-null      int64
 1   CustomerID  4 non-null      int64
 2   Amount      3 non-null      float64
dtypes: float64(1), int64(2)
memory usage: 228.0 bytes
```

In [11]:
```
customers["Name"] = customers["Name"].fillna("Unknown")
```

In [12]:
```
customers["Age"] = customers["Age"].fillna(customers["Age"].mean())
```

In [13]:
```
customers["City"] = customers["City"].fillna("Unknown")
```

In [14]:
```
orders["Amount"] = orders["Amount"].fillna(orders["Amount"].mean())
```

In [15]:
```
customers = customers.drop_duplicates()
orders = orders.drop_duplicates()
```

In [16]:
```
merged = pd.merge(customers, orders, on="CustomerID")
```

In [17]:
```
print("\nMerged Data")
print(merged)
```

```
print("\nMerged Info")
merged.info()

print("\nMerged Describe")
merged.describe()
```

```
Merged Data
   CustomerID       Name   Age     City  OrderID  Amount
0         101  Saptanshu  22.0    Delhi        1   500.0
1         101  Saptanshu  22.0    Delhi        3   550.0
2         104     Saumya  22.0  Unknown        4   450.0
3         102     Hardik  21.0   Mumbai        2   700.0

Merged Info
<class 'pandas.DataFrame'>
RangeIndex: 4 entries, 0 to 3
Data columns (total 6 columns):
 #   Column      Non-Null Count  Dtype
---  ------      --------------  -----
 0   CustomerID  4 non-null      int64
 1   Name        4 non-null      str
 2   Age         4 non-null      float64
 3   City        4 non-null      str
 4   OrderID     4 non-null      int64
 5   Amount      4 non-null      float64
dtypes: float64(2), int64(2), str(2)
memory usage: 324.0 bytes

Merged Describe
```

Out[17]:

	CustomerID	Age	OrderID	Amount
count	4.000000	4.00	4.000000	4.000000
mean	102.000000	21.75	2.500000	550.000000
std	1.414214	0.50	1.290994	108.012345
min	101.000000	21.00	1.000000	450.000000
25%	101.000000	21.75	1.750000	487.500000
50%	101.500000	22.00	2.500000	525.000000
75%	102.500000	22.00	3.250000	587.500000
max	104.000000	22.00	4.000000	700.000000

In [ ]: